

FORECASTING SPOT PRICE OF CRUDE OIL IN CHINA USING RELATIONSHIP WITH FUTURES MARKET

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ABSTRACT

By using the spot price and futures price of crude oil from March, 2018, we found that simple no-change forecasts tend to be more accurate than futures-based models when predicting the spot price of Shengli crude oil in China. Moreover, the spread between futures price and spot price can increase the forecasting accuracy. With the premise of the existence of convenience yield, the inversely proportional relationship between the spot price and the spread can be explained by the change of the sentiment to uncertainty of future crude oil supply. Finally, though the yuan-dominated INE crude oil futures appeared, the Brent and WTI crude oil futures which have been seen as benchmarks for a long time still have better forecasting abilities. This paper takes a look at the INE futures and provides new insights into the Chinese crude oil spot price prediction models comparison.

Keywords: Crude Oil, Spot Price Forecasting Models, Futures Spread

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1. INTRODUCTION

The high occurrence of geopolitics events have driven the volatility of the prices of international crude oil series. For example, the Russia-Ukraine war in 2022 has pushed the Brent oil price to a historical peak of more than \$139.13 per barrel. Since oil price fluctuations have crucial impacts on macroeconomic and financial markets across the world (Basher, et al., 2018; Baumeister and Peersman, 2013; Choi et al., 2018; Hamilton, 1983; Kang et al., 2015; Kilian and Park, 2009; Ma et al., 2018; Sek, 2017), accurately forecasting crude oil price is critical to central banks and oil-related enterprises (Alquist et al., 2013; Safari and Davallou, 2018). Traditional time series models use futures prices to predict spot prices, however, given the unpredictable market movements, whether futures prices are still effective indicators deserved re-examination.

This paper attempts to re-examine several econometrical models, including simple no-change forecast, random walk with drift, futures-based model and spread-based model, to predict the spot price of crude oil for the Chinese market. Using the price series of Medium and Sour crude oil futures traded on Shanghai International Energy Exchange (INE), West Texas Intermediate (WTI) crude oil futures, as well as the Brent crude oil futures, this paper compares the forecasting power of these models from the inception of INE crude oil futures in March, 2018.

As the largest importer and second-largest consumer of crude oil around the world, China's economic development relies heavily on crude oil consumption. According to

the National Bureau of Statistics, from 2018 to 2021, China's crude oil consumption rose from 630.04 million tons to 722.98 million tons, with an increase of about 14.69%. And the crude oil imports increased from 461.89 million tons to 563.99 million tons from 2018 to 2023. Due to the rapid increase and severe volatility, it is necessary to objectively reflect the status of energy supply and demand, provide enterprises and investors engaged in energy production, circulation, and consumption with tools for price discovery and risk management, as well as investment management functions. As a result, Medium and Sour crude oil futures was listed for trading on INE as the first international commodity futures in China on March 26, 2018, aimed to cultivate its own regional or international benchmark. INE crude oil futures has gradually increased in trading activities, with both trading volume and turnover reaching new highs year by year. And the trading volume of INE crude oil futures has ranked third since 2019.

However, it is clear that the construction of China's crude oil futures market had a relatively late beginning and was less efficient than other well-established futures markets (Yang et al., 2020). Furthermore, the launch of the new yuan-denominated oil futures contract has reopened the debate about the relationship between Brent and WTI and the role they play in the oil price discovery process. As the Brent and WTI futures contracts have been chosen as the global oil benchmarks for a long time, the forecasting ability of them and whether they are better predictors should be taken into consideration.

This paper drawn the conclusion of the more accurate prediction ability of spread-based models, and it is also discovered that WTI and Brent crude oil futures, which are considered to be international benchmarks for a long time, can help predict the spot price of Shengli crude oil better than the new set INE futures. However, solely using the futures price of crude oil shows weaker forecasting power than the no-change model, while there is also no need to consider a drift when using the random walk model.

Furthermore, by looking into the spread between the futures and the spot price in detail, the predicting power of spread models can be explained by the uncertainty of the future oil supply. A theoretical general equilibrium model (Alquist and Kilian, 2010) is introduced to explain the relationship between the spread and the spot price. As inventories of crude oil may serve to avoid interruptions of the production process or to meet unexpected shifts in demand, convenience yield is used to reflect the value of keeping the spot crude oil rather than longing the futures. Because of the existence of a marginal convenience yield for crude oil, an increase in uncertainty increases the real spot price and decreases the oil futures spread (Alquist and Kilian, 2010). The changes about the future supply make the spread-based models more accurate than the simple no-change model. The spot price changes due to energy events, such as the outbreak of Covid-19 and Russia-Ukraine War, can proof the relationship between the expectation shifts about the supply and the spot price.

This paper contributes in the following three aspects. The first contribution is to pay attention to the crude oil produced in China. Medium and Sour crude oil are produced in Shengli and Daqing Oilfield in China, while light crude oil is more common in other main Oilfields around the world, such as North Sea Oil Field. The sulfur content of Shengli crude oil is higher than WTI or Brent, which lead to the different pricing mechanism. The second contribution is to examine the forecasting accuracy and discuss the dispute whether simple no-change forecast is the most accurate method by using the evidence from China. The third contribution is to take a look at the yuan dominated crude oil futures traded on INE, and make comparisons with international benchmarks. By using the futures price series to predict the spot price, it implies the difference forecasting ability between INE futures and WTI or Brent futures. It seems that there is still a long way for INE crude oil futures to become a benchmark.

The remaining sections of this paper are structured as follows. Section 2 presents the relevant literatures on the predictors and prediction models of crude oil. Section 3 describes the theoretical model to discuss the relationship between spot price and the spread. Section 4 introduces the spot price forecasting models and performance criteria. Section 5 shows the sources of the futures price and the spot price series, and describes the data briefly. Section 6 shows the empirical results and analyses the forecasting differences between models. Section 7 discusses the spread in detail and introduce the theoretical model. Section 8 concludes the entire article.

2. LITERATURE REVIEW

2.1. The Predictors of Crude Oil Price

Demand and supply fundamentals are considered to decide the spot price of crude oil and are claimed to be major fundamentals based on the analysis of the traditional economics (Kilian, 2009; Kilian, 2014; Hao et al., 2020). Moreover, global economic activity (Kilian, 2009; Wang and Sun, 2017), geopolitical shifts (Zhang et al., 2022; Lyu et al., 2024), macroeconomic (Bouazizi et al., 2024; Zhang et al., 2024) and financial indicators (Fiszeder et al., 2023; Zhang and Wang, 2019) largely affects oil price, and have been integrated into forecasting models to augment the precision of oil price predictions.

However, crude oil is not a general commodity, it is always considered to be an important strategic material, and the financialization of crude oil markets has changed the pricing mechanism. According to the price discovery theory, futures price reflects investors' expectation of spot price in the future. Companies and investors use derivatives, such as futures, as a conventional method to hedge risk, and different futures contracts can reflect the future demand and supply of their underlying assets. The price series of derivatives have been widely used to forecast the spot price of commodities, especially crude oil. Additionally, the futures market is developed on the basis of the spot market, so they have very similar reactions to new market information, and the direction of movement of the futures price and the spot price is basically the same. Previous studies have shown oil futures are highly correlated with oil spot (Alquist and Kilian, 2010; Bai and Li, 2022).

INE crude oil futures has become the world's third-largest crude oil futures market with rapidly growing trading volume and open interest since its launch. Furthermore, the price relationship between INE futures and the spot crude oil has been researched. INE crude oil futures is efficient to convey the fundamental information in crude oil spot market by using cointegration and Granger causality test (Yang et al., 2019). In the long run, the INE crude oil futures' returns have an equilibrium relationship with the spot returns on the Daqing, Shengli, Oman, WTI, and Brent spot markets, imply that the INE crude oil futures price can reflect the fundamental information of spot markets effectively (Yang Chen et al, 2020). By examining the price discovery performance of INE crude oil futures for the spot prices of 19 types of deliverable and non-deliverable Asian crude oil, it is found that the INE crude oil futures market has evolved into a useful and important information source in pricing Asian crudes (Yu et al.,2022). As the price of crude oil futures is widely used to predict the future spot price, and the fact that there are indeed internal relations between the futures price of INE and the spot price. It is natural to believe that the price series of INE crude oil are efficient predictors when forecasting the future price of Chinese crude oil.

Crude oil futures markets are globally dominated by WTI crude oil in America and Brent crude oil in the UK. It is natural to compare the forecasting ability between INE futures and other international benchmarks. Moreover, a major portion of the world's oil trade and long-term oil contracts, including those in Asia, are traditionally based on the WTI and particularly Brent crude oil futures prices (Fattouh, 2011). INE crude oil futures contract is on the path to emerge as an Asian benchmark (Yu et al.,2022).

The oil futures spread is defined as the percent deviation of the oil futures price from the spot price of oil. The spread between the spot price and the futures price can also be used as an effective indicator of whether the price of oil is likely to go up or down (Gramlich, 2004). If the futures price is equal to the expected spot price, the spread between the two should reflect the anticipated change in spot prices (Baumeister and Kilian, 2012).

Current researches focused on the predictors of crude produced in the US or North Sea Oilfield, however, the researches about what determines the price of the Medium and Sour crude oil produced in China are not sufficient. This paper complements the comparisons of the predicting power among futures prices and related statistics such as futures spread.

3. THEORETICAL FRAMEWORK AND DISCUSSION

Alquist and Kilian (2010) represented a theoretical model of the oil futures and oil spot markets, with the conclusion of an increased uncertainty about future oil supply shortfalls causes the spread to fall and the spot price to increase. Such uncertainty shifts raise the current real spot price of oil, as precautionary demand for oil inventories increases in response to increased uncertainty.

In each period, a crude oil importing country chooses next period's inventory holdings of oil (I_t), and the quantity of oil to use in producing the consumption good (Z_t). The oil endowment ($\tilde{w}_t = w_t + \varepsilon_t$) of oil producing countries should equal to the sum of increment of crude oil inventory (ΔI_t) and the quantity to use in the production of the consumption good.

$$\tilde{w}_t = \Delta I_t + Z_t \quad (1)$$

The oil importing country chooses Z_t such that the marginal product of oil equals the real price of oil on terms of consumption good (S_t/P_t).

$$S_t/P_t = F'(Z_t) = F'(\tilde{w}_t - \Delta I_t) \quad (2)$$

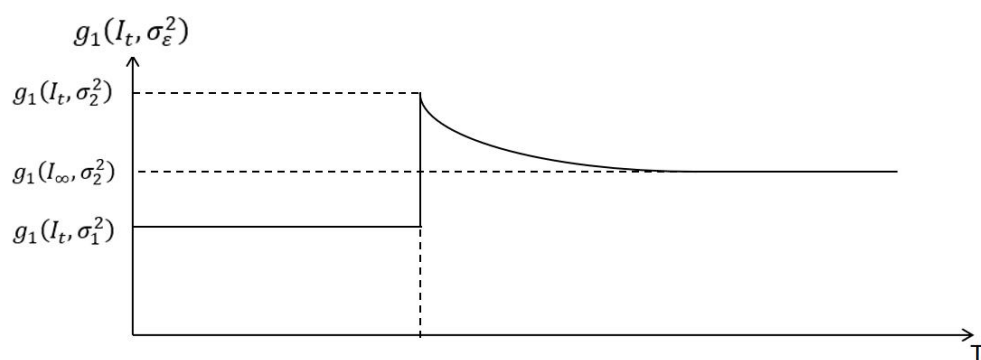
Where S_t denotes the spot price of oil and P_t denotes the numeraire price of the consumption good in period t . $F(Z_t)$ is the production function and $F'(Z_t) > 0, F''(Z_t) < 0$. As ΔI_t increases, the real price of oil will increase correspondingly.

Considering the existence of a market for storage, holding above-ground inventories

instead of having positions in crude oil futures brings benefits, which can be represented as convenience yield $g = g(I_t, \sigma_\varepsilon^2)$. The convenience yield is influenced by the inventory and the variance of oil endowment of oil-producing countries (σ_ε^2) . Let g_I denotes the marginal convenience yield which is the first order derivate of inventories and g_{II} is the cross-partial derivate of g with respect to the argument I_t . It is imposed that $g_I(I_t, \sigma_\varepsilon^2) > 0$ and $g_{II}(I_t, \sigma_\varepsilon^2) < 0$. Moreover, with an increased variance of oil endowment, the marginal convenience yield increases, which means $g_{I2}(I_t, \sigma_\varepsilon^2) > 0$.

When there is an exogenous mean-preserving increase in uncertainty about future oil supply shortfalls, the marginal convenience yield increases immediately. The oil-importing country will hold more inventory because of the decision of an increase of the above-ground inventory in response to the future supply uncertainty. This implies a slow decrease in the marginal convenience yield. Figure 1 illustrates the dynamic change of ΔI_t and $g_I(I_t, \sigma_\varepsilon^2)$. Figure 1(a) plots the change of the increment of inventory and Figure 1(b) shows the corresponding change of marginal convenience yield.

(a)



(b)

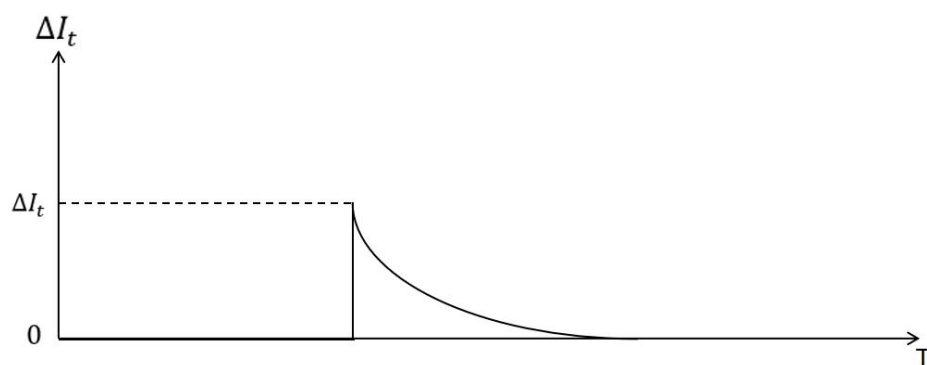


Figure 1. The effect of an increase in uncertainty on ΔI_t and $g_1(I_t, \sigma_\varepsilon^2)$

Note: (a) The effect of an increase in uncertainty on the additional inventory. (b) The effect of an increase in uncertainty on the marginal convenience yield. Time (T) is drawn on the horizontal axis.

Under the premise of satisfying the no-arbitrage conditions in the oil market, the bond market and the market for storage, the returns of storing a barrel of oil for one period and selling short a crude oil futures contract should equal to the return on investing the same amount at risk-free rate $(r_{t,t+1})$.

$$(1 + r_{t,t+1}) g_1(I_t, \sigma_\varepsilon^2) - \frac{S_t}{P_t} + E_t \left[\frac{F_t}{P_{t+1}} \right] = r_{t,t+1} \frac{S_t}{P_t} \quad \#(3)$$

Since the expected real price of the marginal futures contract must equal the expected real price of a barrel of oil next period under the premise of no-arbitrage condition in

futures market, which is given by $E_t \left[\frac{F_t}{P_{t+1}} \right] = E_t \left[\frac{S_{t+1}}{P_{t+1}} \right]$, and using the marginal product to substitute the real price of crude oil, which is given by equation (2), the real price of oil consists of two parts:

$$S_t/P_t = \frac{E_t \left[F'(\tilde{w}_{t+1} - \Delta I_{t+1}) \right] + g_1(I_t, \sigma_\varepsilon^2)}{1 + r_{t,t+1}} \quad \#(4)$$

Due to the existence of convenience yield, the spot price of commodities should larger than the futures price at the same time, the futures spread should be negative during the time period. According to Table 7, the mean value of the spread between INE futures price and spot price is -0.6459, though it is not different from zero at the significant level of 5%. Moreover, there are 40 futures spreads smaller than zero out of all 70 observations. Additionally, the spread between the WTI futures spread and the spot price is statistically significant less than zero. These can prove the existence of convenience yield in Chinese commodity market.

Because the mean-preserving increase of uncertainty cause $E_t \left[F'(\tilde{w}_{t+1} - \Delta I_{t+1}) \right]$ to increase as the marginal product is convex, and it also increase the marginal convenience yield, the real price of oil will jump. As the marginal convenience yield falls because of the slowing rate of additional inventory over time, the real price of oil will decline from its initial peak and come to a new equilibrium, which is higher than its level at the last period. Figure 2 illustrates the dynamic change of the real price of crude oil.

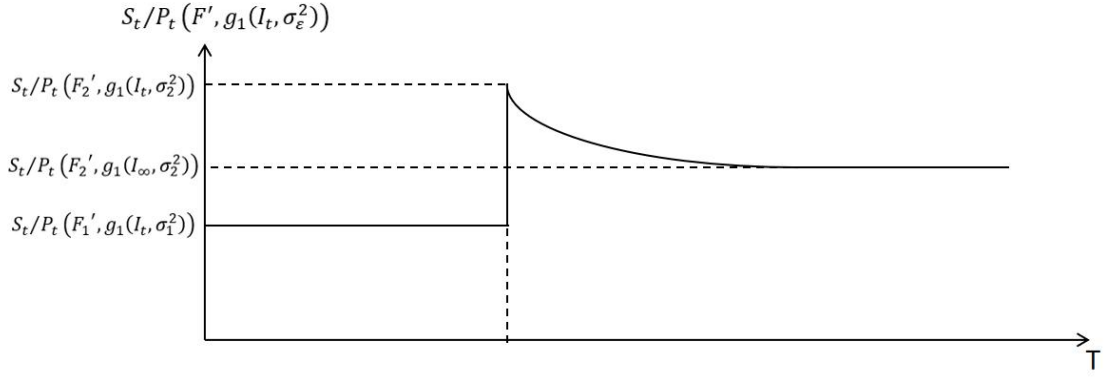


Figure 2. The effect of an increase in uncertainty on S_t/P_t

Note: Time (T) is drawn on the horizontal axis.

Thus, it is clear that the spot price of crude oil will overshoot in response to the increase in uncertainty about future supply shortfalls under the condition of no-arbitrage. To examine the relationship between the increased uncertainty and the futures spread, it is necessary to obtain the expression for the spread. Under the no-arbitrage condition in the bond market, the real value of a bond at time t must equal the discounted real present value of a bond at time $t + 1$, which means $E_t [F_t/P_{t+1}] \approx F_t/P_t$. By rearranging equation (3):

$$(F_t - S_t)/S_t = r_{t,t+1} - \frac{(1 + r_{t,t+1}) g_1(I_t, \sigma_\varepsilon^2)}{S_t/P_t} \#(5)$$

It can be verified that provided g_{12} is large enough, which means the marginal convenience yield is quite sensitive to the increased uncertainty, the first order derivate of futures spread with respect to the uncertainty part of oil endowment (ε_t) is negative.

$$\frac{\partial (F_t - S_t)/S_t}{\partial \varepsilon_t} < 0 \text{ (} g_{12} \text{ is large enough) } \#(6)$$

Hence the negative of the spread may be viewed as an indicator of fluctuations in the spot price of crude oil driven by shifts in precautionary demand for oil, and the spread is in inverse proportion to the spot price of crude oil.

4. FORECASTING MODELS AND PERFORMANCE CRITERIA

4.1. Spot Price Forecasting Models

Oil price forecast studies are mainly based on econometric models, among which autoregressive integrated moving average (ARIMA) (Xiang and Zhuang, 2013), generalized autoregressive conditional heteroskedasticity (GARCH) (Agnolucci, 2009; Mohammadi and Su, 2010), random walk (RW) and vector error correct model (VECM) (Murat and Tokat, 2009) have been widely applied.

Benchmark model

Arguably, a natural benchmark that is commonly used is the simple no-change model, which is also called the naive forecasting method. The no-change model is based on the theoretical assumption that spot prices of commodities are unpredictable. Therefore, the current spot price is the most accurate forecast for the future spot price (Yoon, 1998).

However, whether the simple no-change benchmark model perform better than other existing models has been a controversial problem in the academic field. On the one hand, a growing literature documents that model-based forecasts of the real price of crude oil can outperform a simple no-change benchmark, even over short forecasting horizons (Alquist and Kilian, 2010; Baumeister and Kilian, 2012; Alquist et al., 2013; Baumeister et al., 2014; Baumeister and Kilian, 2014; Chen, 2014; Baumeister and Kilian, 2015; Baumeister et al., 2015; Wang et al., 2015; Baumeister et al., 2018; Funk, 2018; Snudden, 2018; Garratt et al., 2019; Alquist et al., 2020; Baumeister et al.,

2022). On the other hand, at the one-step-ahead prediction, all existing forecasts perform worse than the end-of-month no-change forecast, and this result call into question the usefulness of existing forecasting approaches for the real price of crude oil relative to simple no-change forecasts (Ellwanger and Snudden, 2023).

Let S_t denote the current spot price of oil at time t , and $\widehat{S}_{t+1|t}$ is the expected future spot price at time $t + 1$ conditional on information available at t . The random walk model without drift implies the future spot price of crude oil is simply the current spot price, as the changes are unpredictable. The benchmark model is shown in Equation (7).

$$\widehat{S}_{t+1|t} = S_t \#(\gamma)$$

The following are several forecasting models based on the price of crude oil futures and the spread between the futures price and the spot price.

Random Walk with Drift

One example of econometric models is the random walk without drift introduced above as the Benchmark model. Given that oil prices have been persistently trending upward or downward at times, it is natural to consider a random walk model with drift. The drift in the forecasting model below is estimated recursively.

$$\widehat{S}_{t+1|t} = S_t \left(1 + \widehat{\alpha} \right) \#(\vartheta)$$

Forecasts Based on the Futures Prices

It is common that using the price of futures as a great available predictor to forecast the future spot price at maturity. Let F_t denote the price of crude oil futures contracts at time t .

$$\widehat{S}_{t+1|t} = F_t \quad \#(9)$$

Forecasts Based on the Futures Spread

The oil futures spread is defined as the percent deviation of the oil futures price from the spot price of oil, which can also be used as an effective indicator of whether the price of oil is likely to go up or down (Gramlich, 2004). If the futures price equals to the expected future spot price, which means $F_t = E[S_{t+1}]$, the spread should be an effective predictor of the expected change in spot prices. The rationale is from dividing $F_t = E[S_{t+1}]$ by S_t , which results in $F_t/S_t = E[S_{t+1}]/S_t$. Therefore, a well-documented method to forecast future spot prices is to use the spread between the futures and spot prices, and the logarithmic spread can be added into the forecasting model as the change rate of spot price. However, there is no advantage when using the futures spread model to predict the spot price of crude oil and the mean squared prediction errors (MSPE) is relatively larger than the simple no-change model by the horizon of one month (Alquist and Kilian, 2010). Whether the spread is a useful indicator when predicting the crude oil produced in China deserved verification and examination. The baseline model is shown in Equation (10).

$$\widehat{S}_{t+1|t} = S_t \left(1 + \ln \left(F_t / S_t \right) \right) \quad \#(10)$$

An alternative forecasting model is considered to explore the forecasting accuracy of the spread. As the spread may be a biased predictor, the assumption of a zero intercept should be relaxed. Additionally, the proportionality restriction should also be relaxed.

$$\widehat{S}_{t+1|t} = S_t \left(1 + \widehat{\alpha} + \widehat{\beta} \ln (F_t / S_t) \right) \quad (17)$$

Here $\widehat{\alpha}$ and $\widehat{\beta}$ are OLS estimates obtained in real time form recursive regressions.

Table 1. Forecasting Models

Model	Formula
Random walk without drift (Benchmark Model)	$\widehat{S}_{t+1 t} = S_t$
Random walk with drift	$\widehat{S}_{t+1 t} = S_t (1 + \widehat{\alpha})$
Futures prices model	$\widehat{S}_{t+1 t} = F_t$
Futures spread model	$\widehat{S}_{t+1 t} = S_t (1 + \ln (F_t / S_t))$ $\widehat{S}_{t+1 t} = S_t (1 + \widehat{\alpha}$ $+ \widehat{\beta} \ln (F_t / S_t))$

4.2. Performance Criteria

Since forecasting performances of the models have various criteria, and no single measure can fully capture the characteristics of prediction errors. Therefore, four methods are considered to measure the differences of forecasting abilities in this paper, which are the mean squared prediction errors (MSPE), bias, the mean absolute prediction error (MAPE) and the success ratio (Pesaran and Timmermann, 1992). The

bias is defined as the average amount by which the true value of spot price at maturity exceeds the prediction. The success ratio is defined as the fraction of forecasts that correctly predict the sign of the change in the price of oil. The formulations of MSPE, bias and MAPE measures are shown in equations as follows.

$$MSPE = \frac{1}{n} \sum_{i=1}^n \left(\widehat{S}_{t+1|t} - S_{t+1} \right)^2 \quad \#(12)$$

$$Bias = \frac{1}{n} \sum_{i=1}^n \left(\widehat{S}_{t+1|t} - S_{t+1} \right) \quad \#(13)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \widehat{S}_{t+1|t} - S_{t+1} \right| \quad \#(14)$$

Where $\widehat{S}_{t+1|t}$ is the predicted future spot price at time t+1, conditional on information available at time t, while S_{t+1} is the true spot price of crude oil at time t+1. N is the number of observations.

In addition to the accuracy measures, the statistical significance of two forecasting models should be evaluated properly. Comparisons of non-nested models without estimated parameters (futures-based forecasting model and futures spread based forecasting model without estimated parameters) are based on the DM-test of Diebold and Mariano (2002) with N (0,1) critical values. The DM test employs MSPE as the loss function and assumes the null hypothesis that the MSPE of the tested model is not lower than that of the benchmark model. While, nested model comparisons with estimated parameters (futures spread forecasting model with two parameters) are based on Clark and West (2006).

5. DATA AND VARIABLES

This study uses two types of price series to examine the abilities of several spot price forecasting models from March, 2018 to December, 2023, which are collected from WIND database.

Firstly, this paper uses the spot price of crude oil produced in Shengli Oilfield. There are several main crude oil producers all around China, such as Daqing Oilfield, Changqing Oilfield and Bohai Oilfield. However, only the crude oil produced in Shengli Oilfield is accepted at the Shanghai International Energy Exchange as the underlying asset of Medium and Sour crude oil futures. To obtain the monthly series and correspond to the futures price series, the spot prices of the last trading day each month are selected in the following research.

Secondly, this study uses futures prices of continuous contracts, measured by the daily closing prices of futures contract expiring next month. Medium and Sour Crude Oil (SC) futures traded on INE, Brent futures traded on International Petroleum Exchange (IPE) and daily prices of light crude oil futures traded on the New York Mercantile Exchange (NYMEX) are used as predictors. To gain the monthly data from daily prices, the close prices of continuous contracts of the last trading day in each month are chosen, instead of the monthly average price. Relative to the end-of month no-change forecast, the monthly average no-change forecast is an ineffective predictor (Ellwanger and Snudden, 2023).

There are 70 monthly observations from March, 2018 to December, 2023. Only INE futures are denominated in China's Renminbi, while the spot crude oil and other international futures contracts are dominated in US dollars. To unify the units of two price series, the spot exchange rates between US dollar against RMB from WIND database are used. Table 2 reports the summary statistics for both the spot prices and futures prices.

Table 2. Description of spot prices and futures prices of crude oil

Variable	Obs	Mean	Std dev	Min	Max
Spot Price: Crude Oil (Shengli China)	70	71.07	20.22	24.46	122.55
Futures Price: Medium & Sour Crude oil	70	70.34	18.86	31.98	114.58
Futures Price: IPE Brent Crude Oil	70	72.17	20.04	22.71	122.84
Futures Price: NYMEX Light Crude Oil	70	66.72	19.25	19.09	115.26

Notes: This table reports the basic information about monthly spot prices of Shengli crude oil and monthly futures prices of three crude oil futures from March, 2018 to December, 2023. All the series are denominated in US dollars. Obs denotes observations and Std dev denotes standard deviations of the price series.

According to Figure 3, the closing price of Shanghai Medium and Sour crude oil futures was \$73.25 per barrel as of the end of December 2023, Brent crude oil futures closed at \$77.08 per barrel and light crude oil futures traded on NYMEX closed at \$71.33 per barrel. Additionally, Figure 3 depicts that the price series of spot crude oil in China and the futures prices show strong co-movements during the time period.

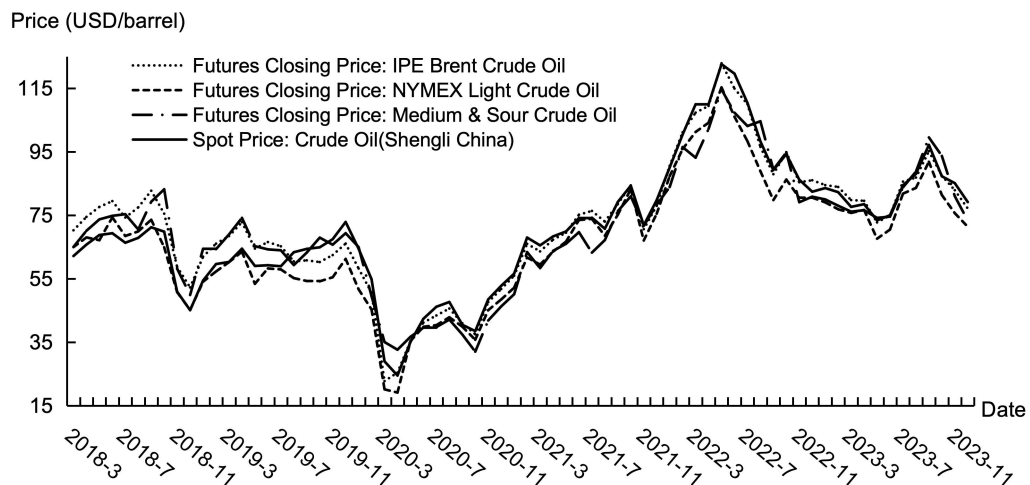


Figure 3. Crude oil futures prices and spot prices from 2018.3 to 2023.12

Note: Monthly prices of crude oil futures contracts and spot price of crude oil from March, 2018 to December, 2023 from WIND database.

6. EMPIRICAL RESULTS

Tables in this section assess the predictive accuracy of various forecasting models against the benchmark of a random walk without drift for horizons of 1 month. The forecast evaluation period is from March 2018 to December 2021. The assessment of which forecasting model is most accurate may depend on the loss function of the forecaster (Elliott and Timmermann, 2008). The results for the mean squared prediction error (MSPE) and the mean absolute prediction error (MAPE) are presented in the table. The bias of the forecasts is also reported. The number of times that a forecast correctly predicts the sign of the change in the spot price based on the success ratio statistic of Pesaran and Timmermann (1992) are also presented. In addition to ranking forecasting models by each loss function, the null that a given candidate forecasting model is as accurate as the random walk without drift against the alternative that the candidate model is more accurate than the no-change forecast is formally tested.

6.1. INE Futures Prices as Predictors of Crude Oil Spot Prices

The results in Table 3 show that allowing for a drift in no case significantly lowers either MSPE or MAPE of the random walk model when the drift is estimated based on rolling regressions, but the results are not statically significant. The random walk with drift forecasting model has systematically negative bias, and the absolute value is larger than the benchmark. The random walk with drift model generates forecasts in one direction only, so the PT statistic in Pesaran-Timmermann Test will be undefined. As a result, when forecasting the spot price of crude oil using spot price series as

predictor, it is not necessary to consider a drift.

As to the forecasting ability comparison between futures-based model and the simple no-change forecasts in Table 3, it is documented that the no-change forecast has lower MSPE and MAPE, though the p-values are large. Moreover, the random walk forecast is less biased than the futures forecast, and the success ratio is insignificant at the 5% level. Using the INE futures price to predict the future spot price shows no advantages. The reason why futures-based model is less accurate will be discussed in Section 6.

Alternatively, recursively estimated futures spread regressions may be used to generate forecasts of the price of oil. Table 3 documents the predicting models including the futures spread without and with estimated parameters respectively. It shows that the no-change forecast has lower MSPE and MAPE than spread-based forecasts, but the p-values are large and the results are not statically significant. The absolute values of bias are larger than benchmark models, and the success ratios are not significant. The MSPE and MAPE rankings are still larger than no-change model after relaxing the restrictions and adding two estimated parameters, and the MAPE is statically significant.

Table 3. Recursive forecast error diagnostics (using SC futures price, INE)

$\widehat{S}_{t+h t}$	MSPE (p-value)	Bias	MAPE (p-value)	Success ratio (p-value)
S_t	52.7684	0.2464	5.6116	n.a.
$S_t (1 + \widehat{\alpha})$	1.0434 (0.5355)	-1.1708	1.0348 (0.5195)	0.4348 (n.a.)
F_t	1.1737 (0.7755)	0.3726	1.0400 (0.6200)	0.5000 (0.5000)
$S_t (1 + \ln (F_t / S_t))$	1.1618 (0.7684)	0.4239	1.0331 (0.6015)	0.5000 (0.5000)
$S_t (1 + \widehat{\alpha} + \widehat{\beta} \ln (F_t / S_t))$	1.0457 (0.5626)	-1.0141	1.0260** (0.0300)	0.4348 (n.a.)

Note: The spot prices of crude oil are the price series in Shengli producer and futures prices are the price series of Medium and Sour crude oil futures traded on INE. The forecast evaluation period is from January 2022 to December 2023 and the initial estimation window is from March 2018 to December 2021. All MSPE and MAPE results are presented as ratios relative to the benchmark no-change forecast model, for which the actual level of the MSPE and MAPE are reported. The bias is defined as the average amount by which S_{t+1} exceeds the prediction. p-values are in parentheses. All p-values refer to pairwise tests of the null of a random walk without drift. Comparisons of non-nested models without estimated parameters are based on the DM-test of Diebold and Mariano (2002) with N (0,1) critical values. Nested model comparisons with estimated parameters are based on Clark and West (2006). For the rolling regression estimates of the random walk with drift we use N (0,1) critical values under quadratic loss; for recursive estimates under quadratic loss and for all estimates under absolute loss we use bootstrap critical values as described in Clark and West. The success ratio is defined as the fraction of forecasts that correctly predict the sign of the change in the price of oil. The sign test in the last column is based on Pesaran and Timmermann (1992). *, ** and *** denote rejecting the null hypothesis at 10%, 5% and 1% significant levels, respectively.

6.2. Brent and WTI Futures Prices as Predictors of Crude Oil Spot Prices

Table 4 and Table 5 compare the forecast error results of the futures models against the benchmark model when using the prices of Brent futures and WTI futures respectively. It is clear that the MSPE and MAPE are relatively large though the results are not statistically significant. The absolute values of biases are larger than no-change forecasts and the futures forecasts have no advantages when it comes to predicting the sign of the change in oil prices. As a result, Brent and WTI futures prices are no more accurate predictors of the price of crude oil than simple no-change forecasts over this period.

As to the prediction power comparison between no-change model and spread-based models, Row 3 in each table shows the prediction power between no-change model and spread-based model without estimated parameters and Row 4 compares the spread model with relaxed restrictions with the Benchmark.

According to the four performance criteria, the results show that no-change forecasts are also more accurate. It is noteworthy that the spread model based on the futures prices in NYMEX generates forecasts in one direction only. Without adding estimated parameters, the spreads between futures prices and the spot prices are not more accurate. Moreover, it is clear that MSPE is lower than benchmark model though the p-value is large. While the MAPE is lower as well and it is statically significant at the 5% level. The no-change forecast is less accurate than futures spread based model based on the MSPE and MAPE rankings, when using the futures spreads of Light

crude oil traded on NYMEX as predictors. Additionally, the MSPE and MAPE results are statically significant at the 10% level. The absolute values of biases of these models show that no-change model is better. When it comes to success ratio, the spread models show no advantages.

The analyses of these forecasting models reveal that using spread-based models consisting of estimated parameters can help predict the spot price of crude oil better than the simple no-change model when using Brent and WTI futures spreads. This conclusion will be discussed in detail in Section 6.

Table 4. Recursive forecast error diagnostics (using Brent futures price, IPE)

$\widehat{S}_{t+h t}$	MSPE (p-value)	Bias	MAPE (p-value)	Success ratio (p-value)
S_t	52.7684	0.2464	5.6116	n.a.
F_t	1.0042 (0.5125)	-0.6625	1.0379 (0.6617)	0.4167 (0.7296)
$S_t (1 + \ln (F_t / S_t))$	1.0013 (0.5038)	-0.6456	1.0349 (0.6503)	0.4167 (0.7296)
$S_t (1 + \widehat{\alpha} + \widehat{\beta} \ln (F_t / S_t))$	0.9800 (0.1071)	-0.4768	0.9893** (0.0490)	0.5217 (0.3814)

Note: The spot prices of crude oil are the price series in Shengli producer and futures prices are the price series of Brent crude oil futures traded on IPE. *, ** and *** denote rejecting the null hypothesis at 10%, 5% and 1% significant levels, respectively.

Table 5. Recursive forecast error diagnostics (using Light futures price, NYMEX)

$\widehat{S}_{t+h t}$	MSPE (p-value)	Bias	MAPE (p-value)	Success ratio (p-value)
S_t	52.7684	0.2464	5.6116	n.a.
F_t	1.4748 (0.8416)	4.1842	1.2029 (0.8347)	0.5000 (n.a.)
$S_t (1 + \ln (F_t / S_t))$	1.5198 (0.8588)	4.3377	1.2302 (0.8673)	0.5000 (n.a.)
$S_t (1 + \widehat{\alpha} + \widehat{\beta} \ln (F_t / S_t))$	0.9043* (0.0581)	-1.1273	0.9761* (0.0601)	0.5217 (0.2755)

Note: The spot prices of crude oil are the price series in Shengli producer and futures prices are the price series of light crude oil futures traded on NYMEX. *, ** and *** denote rejecting the null hypothesis at 10%, 5% and 1% significant levels, respectively.

7. FORECASTING MODEL BASED ON FUTURES AND SPREAD

7.1. Why Futures-Based Model is Less Accurate

The results above show that the futures-based model has no advantages than simple no-change model when predicting the spot price, no matter which futures price series are used. According to the official website of Shanghai International Energy Exchange, nine types of crude oil are accepted at the maturity, including UAE Dubai crude oil, UAE Upper Zakum crude oil, UAE's Murban crude oil, Oman crude oil, Qatar Marine crude oil, Iraq's Basrah light, Iraq's Basrah medium, Brazil's Tupi crude oil and China's Shengli crude oil. Most deliverable oil types for other main crude oil futures are local crude oil. For example, Brent crude oil futures accept four types of light sweet crude oil (Brent, Forties, Oseberg, Ekofisk and Troll) produced from the North Sea Oil Field. According to the public data of stock on warrant of SC futures traded on INE in total from October 26, 2018 to November 26, 2023 (see Figure 4), most of the crude oil for delivery is from the Middle East and focus on Oman crude oil (Oman), Upper Zakum crude oil (Arabic) and Batura light oil (Iraq).

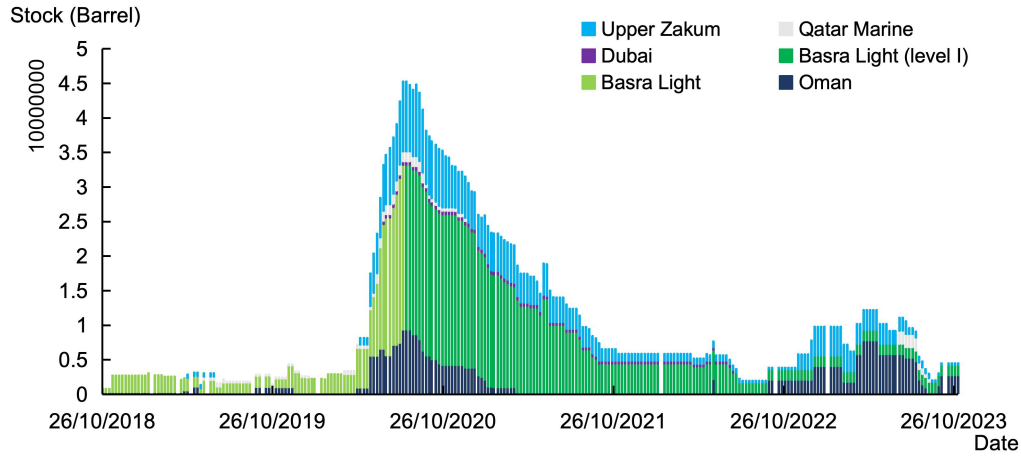


Figure 4. INE crude oil futures' closing stock on warrant.

As one of the major contributions of the futures market are risk transfer and price discovery (Garbade and Silber, 1983). Even though the Shengli crude oil is acceptable at INE, there are almost no stock on warrant since March, 2018. While the Brent futures contract and WTI futures contract do not accept crude oil produced in China in delivery. It is natural that the futures traded in different exchanges around the world cannot diversify the risk of crude oil produced in Shengli Oilfield. As a result, it is plausible that the futures price series are not effective predictors when forecasting the future spot price, and no-change forecasts tend to be more accurate.

7.2. Relationship with Forecast Efficiency Regressions

A test to show whether the spread is a great predictor by testing whether the risk premium is time-varying is used to confirm the inaccuracy of using futures price or spread to predict the spot price (Chernenko et al., 2004; Chinn et al., 2005). Consider the full-sample regression model:

$$\ln(S_{t+1}/S_t) = \alpha + \beta \ln(F_t/S_t) + \varepsilon_t \quad (15)$$

where ε_t denotes the error term. Forecast efficiency in the context of the oil futures spread means that the hypothesis: $H_0: \alpha = 0, \beta = 1$ holds. If the null hypothesis can be accepted, it means the spread can be used as the indicator of the change rate of spot price. This is often known as the Mincer-Zarnowitz regression (Mincer and Zarnowitz, 1969). If the slope coefficient β is significantly different from 1, it can be concluded that the risk premium is time-varying (Fama and French, 1987, 1988; Chernenko et al., 2004), and using the spread as the predictor is not suitable.

For the full sample, however, the hypothesis of forecast efficiency ($\hat{\alpha} = 0, \hat{\beta} = 1$) is rejected at the 5% level (see Table 6), no matter which futures price is used. The estimated slope coefficient β is much smaller than 1, and the null hypotheses of $\hat{\beta} = 1$ are rejected. This means the forecast of the conditional mean is not correct. This pattern reveals that the forecasting models based on spread before relaxing the assumption of a zero intercept and the proportionality restriction, are not more efficient compared to the no-change model, which are consistent with the results above.

Table 6. Asymptotic p-values for forecast efficiency regressions

Futures price series	$\hat{\alpha}$	$\hat{\beta}$	$H_0: \hat{\alpha} = 0$	$H_0: \hat{\beta} = 1$	$H_0: \hat{\alpha} = 0, \hat{\beta} = 1$
INE	0.0039	0.4019	0.0727	0.0001	0.0006
Brent	-0.0020	0.4539	0.3670	0.0139	0.0346
WTI	0.0297	0.0161	0.3876	0.0022	0.0000

Note: All t-and Wald-tests have been computed based on HAC standard errors.

7.3. Describe the Spreads in Great Detail

According to the analysis above, adding parameters α and β into the model based on spread can make the forecasting process more accurate. It is necessary to examine the spread between the futures price and the spot price in greater detail. For this purpose, it is convenient to express the deviation in percentage terms as $\ln(F_t/S_t)$.

In Table 7, using heteroskedasticity and autocorrelation consistent (HAC) standard errors, shows Spread₁ (the spread between the price of SC futures and the spot price) and Spread₂ (the spread between Brent futures price and the spot price) are not statistically different from zero on average at the 5% level. Spread₃ (the spread between light crude oil futures traded on NYMEX and spot price) is statistically smaller than 0. All of the mean absolute deviations of three spreads are comparatively large. At any point in time, the discrepancy between the futures price and the spot price may be very large and go in either direction. For example, if Spread₁ is 0 today, the spread between the logarithmic form of INE futures price and the logarithmic form of spot oil price can be as large as 28.70% or as small as -18.45%. It is this variability of the deviation of futures prices from spot prices rather than the differences in mean that drives the larger MSPE of futures-based forecasts and that makes the use of such oil price forecasts inadvisable.

Table 7. Time series features of $\ln(F_t/S_t)$ (%)

	Spread ₁	Spread ₂	Spread ₃
Mean	-0.6459	1.4514	-6.7961
(p-value)	(0.3740)	(0.1935)	(0.0000)
Mean absolute deviation	7.7340	5.2396	7.5152
Maximum	28.7006	15.0027	6.9676
Minimum	-18.4545	-24.1728	-36.3814

Note: Spread₁ is the spread between the price of SC futures and the spot price. Spread₂ is the spread between Brent futures price and the spot price. Spread₃ is the spread between light crude oil futures traded on NYMEX and spot price. All spreads are in percentage terms. The p-values of the test for a zero mean are based on HAC standard errors.

According to Figure 5, the trend of Spread₂ (the spread between Brent futures price and the spot price) is similar to that of Spread₃ (the spread between light crude oil futures traded on NYMEX and spot price), while Spread₁ (the spread between the price of SC futures and the spot price) shows a different trend.

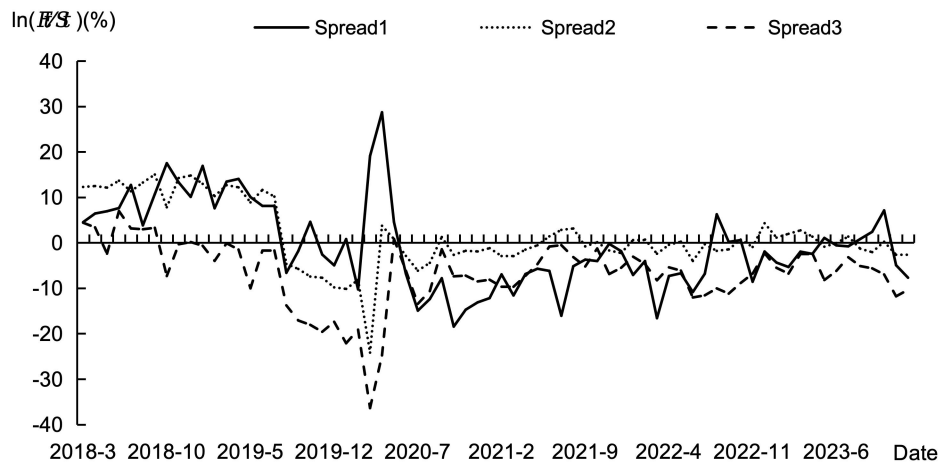


Figure 5. The real value of $\ln(F_t/S_t)$ (%) from 2018.3 to 2023.12

Note: Spread₁ is the spread between the price of SC futures and the spot price. Spread₂ is the spread between Brent futures price and the spot price. Spread₃ is the spread between light crude

oil futures traded on NYMEX and spot price. All spreads are in percentage terms.

7.4. Test the Moves of Spread during Known Periods of Uncertainty Shifts

Verifying the futures spread moved in the expected direction at times of major events, such as the outbreak of wars, which have an impact on the shifts of uncertainty about future crude oil supply, is one direct way of judging the conclusions Figure 4 plots the spreads and the spot price of crude oil from March, 2018 to December, 2023 and focuses on the attack to oil facilities in Saudi Arabia (which should have caused the spread to fall), Covid-19 (which should have caused the spread to increase as world demand for crude oil fell, making a shortfall less likely) and Russia-Ukraine War (which should have caused the spread to fall). Expectations about the future supply shifts are not the only possible reason for changes in the futures spread, but arguably they are the most important reasons.

Two oil facilities in Saudi Arabia were attacked in the early morning of September 14, 2019, and reduced Saudi Arabia's oil supply by 5.7 million barrels per day, exceeding half of its oil production at that time. The spread declined in response to the increased uncertainty about future oil supply. However, with Saudi Arabia immediately using its oil inventory and the rapid recovery of oil supply capacity, international oil prices quickly fell back, and the volatility of futures spread was not that severe.

The pandemic outbreaked in December, 2019, and spread around the globe at the start of 2020. This was a historical moment in the history of human civilization and has a

profound influence on the international economy. The world demand of crude oil fell suddenly, making the shortfall less likely in the near future. The pandemic should cause the spread to increase, and the spot price of crude oil will decrease correspondingly. According to Figure 4, all series fluctuated severely in the year of 2020. The outbreak of pandemic decreased the uncertainty as world demand for crude oil fell, the spread surged and the spot price decreased.

At the beginning of 2022, with the outbreak of the war between Russia and Ukraine, the price of crude oil surged as Russia was one of the largest oil producing countries. With the increase of uncertainty of oil supply, the spread should decrease and the spot price will increase. During 2022, though the crude oil produced in Russia exports to America has been halted, the exports to other countries and regions were increasing, the spread just increased slightly. It confirms the analysis that an increase in uncertainty decreases the spread and increases the spot price.

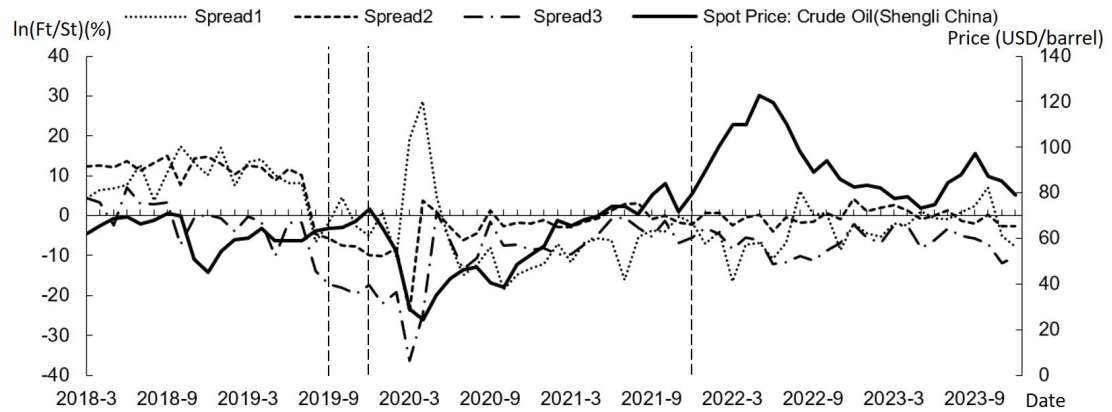


Figure 6. The spread between futures price and spot price.

Note: Spread1 is the spread between the price of SC futures and the spot price. Spread2 is the spread between Brent futures price and the spot price. Spread3 is the spread between light crude oil futures traded on NYMEX and spot price. All spreads are in percentage terms. The spot prices are on the left axis and the spreads are on the right axis. The vertical lines mark the attack to the oil facilities in Saudi Arabia, the outbreak of COVID-19 and Russia-Ukraine War.

However, it is possible that several international events happened and influenced the price of crude oil to varying degrees at the same time. The events above are not the only possible triggers of the shifts of the spread, but it is plausible to consider them as the most important ones.

8. CONCLUSION

This paper provides new insights on the interpretation of crude oil futures prices and the futures spread. And contribute to the selection of efficient predictors and accurate forecasting models of the spot price of crude oil in China. The key points can be summarized as follows.

Firstly, using observations since the first trading day of INE futures, it is clear that the futures prices are not the most accurate predictors of the spot price in practice. Instead, forecasting models based on the spread between futures price and spot price with two estimated parameters tend to be more accurate. And the spreads between Brent or WTI futures and spot price of Shengli crude oil are better predictors than the spread calculated using INE futures price. Additionally, no-change forecasts are also more accurate than random walk with drift models.

Secondly, the disadvantages of future-based forecasting model can be explained by the restrictions of deliverable crude oil. Additionally, it is not suitable to use the spread as the predictor of the change rate of spot price according to Mincer-Zarnowitz regression. By looking at the spread in detail, the discrepancy between the futures price and the spot price may be very large and go in either direction. It is this variability of the deviation rather than the differences in mean that drives the larger MSPE of futures-based forecasts.

Finally, to explain the better forecasting ability of spread-based models, a theoretical general equilibrium model was introduced. Increased uncertainty about future oil

supply shortfalls causes the spread to fall, which raise the current real spot price of oil due to the existence of convenience yield. As the marginal convenience yield is quite sensitive to the increased uncertainty, the negative of the spread may be viewed as an indicator, and the spread is in inverse proportion to the spot price of crude oil. Several international energy-related events, such as Russia-Ukraine War, can verify these conclusions of the theoretical model.

REFERENCES

- Agnolucci, P. (2009). Volatility in crude oil futures: A comparison of the predictive ability of GARCH and implied volatility models. *Energy Economics*, 31(2), 316-321.
- Alquist, R., & Kilian, L. (2010). What do we learn from the price of crude oil futures? *Journal of Applied econometrics*, 25(4), 539-573.
- Alquist, R., Kilian, L., & Vigfusson, R. J. (2013). Forecasting the price of oil. In *Handbook of economic forecasting* (Vol. 2, pp. 427-507). Elsevier.
- Bai, L., Li, X., Wei, Y., & Wei, G. (2022). Does crude oil futures price really help to predict spot oil price? New evidence from density forecasting. *International Journal of Finance & Economics*, 27(3), 3694-3712.
- Basher, S. A., Haug, A. A., & Sadorsky, P. (2018). The impact of oil-market shocks on stock returns in major oil-exporting countries. *Journal of International Money and Finance*, 86, 264-280.
- Baumeister, C., & Hamilton, J. D. (2019). Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks. *American Economic Review*, 109(5), 1873-1910.
- Baumeister, C., & Kilian, L. (2012). Real-time forecasts of the real price of oil. *Journal of business & economic statistics*, 30(2), 326-336.
- Baumeister, C., & Kilian, L. (2014). A general approach to recovering market expectations from futures prices with an application to crude oil.
- Baumeister, C., & Kilian, L. (2016). Forty years of oil price fluctuations: Why the price of oil may still surprise us. *Journal of Economic Perspectives*, 30(1), 139-160.
- Baumeister, C., & Peersman, G. (2013). Time-varying effects of oil supply shocks on the US economy. *American Economic Journal: Macroeconomics*, 5(4), 1-28.
- Baumeister, C., Guérin, P., & Kilian, L. (2015). Do high-frequency financial data help forecast oil prices? The MIDAS touch at work. *International Journal of Forecasting*, 31(2), 238-252.
- Baumeister, C., Kilian, L., & Lee, T. K. (2014). Are there gains from pooling real-time oil price forecasts? *Energy Economics*, 46, S33-S43.

- Bouazizi, T., Guesmi, K., Galariotis, E., & Vigne, S. A. (2024). Crude oil prices in times of crisis: The role of Covid-19 and historical events. *International Review of Financial Analysis*, 91, 102955.
- Chen, S. S. (2014). Forecasting crude oil price movements with oil-sensitive stocks. *Economic Inquiry*, 52(2), 830-844.
- Chernenko, S., Schwarz, K., & Wright, J. H. (2004). The information content of forward and futures prices: Market expectations and the price of risk. Available at SSRN 560386.
- Chinn, M. D., LeBlanc, M., & Coibion, O. (2005). The predictive content of energy futures: an update on petroleum, natural gas, heating oil and gasoline.
- Choi, S., Furceri, D., Loungani, P., Mishra, S., & Poplawski-Ribeiro, M. (2018). Oil prices and inflation dynamics: Evidence from advanced and developing economies. *Journal of International Money and Finance*, 82, 71-96.
- Clark, T. E., & West, K. D. (2006). Using out-of-sample mean squared prediction errors to test the martingale difference hypothesis. *Journal of econometrics*, 135(1-2), 155-186.
- Diebold, F. X., & Mariano, R. S. (2002). Comparing predictive accuracy. *Journal of Business & economic statistics*, 20(1), 134-144.
- Elliott, G., & Timmermann, A. (2008). Economic forecasting. *Journal of Economic Literature*, 46(1), 3-56.
- Ellwanger, R., & Snudden, S. (2023). Forecasts of the real price of oil revisited: Do they beat the random walk? *Journal of Banking & Finance*, 154, 106962.
- Fama, E. F., & French, K. R. (1987). Commodity Futures Prices: Some Evidence on Forecast Power, Premiums, and the Theory of Storage. *Journal of Business*, 55-73.
- Fama, E. F., & French, K. R. (1988). Business cycles and the behavior of metals prices. *The Journal of Finance*, 43(5), 1075-1093.
- Fattouh, B. (2011). An anatomy of the crude oil pricing system. Oxford institute for energy studies.
- Fiszeder, P., Fałdziński, M., & Molnár, P. (2023). Attention to oil prices and its impact on the oil, gold and stock markets and their covariance. *Energy Economics*, 120, 106643.
- Gramlich, E. M. (2004). Oil shocks and monetary policy. Speech presented at the Annual Economic Luncheon, Federal Reserve Bank of Kansas City.

- Hamilton, J. D. (1983). Oil and the macroeconomy since World War II. *Journal of political economy*, 91(2), 228-248.
- Hao, X., Zhao, Y., & Wang, Y. (2020). Forecasting the real prices of crude oil using robust regression models with regularization constraints. *Energy Economics*, 86, 104683.
- Kang, W., Ratti, R. A., & Yoon, K. H. (2015). The impact of oil price shocks on the stock market return and volatility relationship. *Journal of International Financial Markets, Institutions and Money*, 34, 41-54.
- Kilian, L. (2009). Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American economic review*, 99(3), 1053-1069.
- Kilian, L., & Murphy, D. P. (2014). The role of inventories and speculative trading in the global market for crude oil. *Journal of Applied econometrics*, 29(3), 454-478.
- Kilian, L., & Park, C. (2009). The impact of oil price shocks on the US stock market. *International economic review*, 50(4), 1267-1287.
- Lyu, Y., Qin, F., Ke, R., Yang, M., & Chang, J. (2024). Forecasting the VaR of the crude oil market: A combination of mixed data sampling and extreme value theory. *Energy Economics*, 133, 107500.
- Ma, F., Liu, J., Wahab, M. I. M., & Zhang, Y. (2018). Forecasting the aggregate oil price volatility in a data-rich environment. *Economic Modelling*, 72, 320-332.
- Mincer, J. A., & Zarnowitz, V. (1969). The evaluation of economic forecasts. In *Economic forecasts and expectations: Analysis of forecasting behavior and performance* (pp. 3-46). NBER.
- Mohammadi, H., & Su, L. (2010). International evidence on crude oil price dynamics: Applications of ARIMA-GARCH models. *Energy Economics*, 32(5), 1001-1008.
- Murat, A., & Tokat, E. (2009). Forecasting oil price movements with crack spread futures. *Energy Economics*, 31(1), 85-90.
- Pesaran, M. H., & Timmermann, A. (1992). A simple nonparametric test of predictive performance. *Journal of Business & Economic Statistics*, 10(4), 461-465.
- Safari, A., & Davallou, M. (2018). Oil price forecasting using a hybrid model. *Energy*, 148, 49-58.
- Sek, S. K. (2017). Impact of oil price changes on domestic price inflation at disaggregated levels: Evidence from linear and nonlinear ARDL modeling. *Energy*, 130, 204-217.

- Snudden, S. (2018). Targeted growth rates for long-horizon crude oil price forecasts. *International Journal of Forecasting*, 34(1), 1-16.
- Wang, K. H., Su, C. W., & Umar, M. (2021). Geopolitical risk and crude oil security: A Chinese perspective. *Energy*, 219, 119555.
- Wang, Q., & Sun, X. (2017). Crude oil price: Demand, supply, economic activity, economic policy uncertainty and wars—From the perspective of structural equation modelling (SEM). *Energy*, 133, 483-490.
- Xiang, Y., & Zhuang, X. H. (2013). Application of ARIMA model in short-term prediction of international crude oil price. *Advanced Materials Research*, 798, 979-982.
- Yang, C., Lv, F., Fang, L., & Shang, X. (2020). The pricing efficiency of crude oil futures in the Shanghai International Exchange. *Finance Research Letters*, 36, 101329.
- Yang, L. (2019). Connectedness of economic policy uncertainty and oil price shocks in a time domain perspective. *Energy Economics*, 80, 219-233.
- Yang, Y., Ma, Y. R., Hu, M., Zhang, D., & Ji, Q. (2021). Extreme risk spillover between Chinese and global crude oil futures. *Finance Research Letters*, 40, 101743.
- Yoon, G. (1998). Forecasting with structural change: why is the random walk model so damned difficult to beat? *Applied Economics Letters*, 5(1), 41-42.
- Yu, Z., Yang, J., & Webb, R. I. (2023). Price discovery in China's crude oil futures markets: An emerging Asian benchmark? *Journal of Futures Markets*, 43(3), 297-324.
- Zhang, L., Chen, Y., & Bouri, E. (2024). Time-varying jump intensity and volatility forecasting of crude oil returns. *Energy Economics*, 129, 107236.
- Zhang, Y. J., & Wang, J. L. (2019). Do high-frequency stock market data help forecast crude oil prices? Evidence from the MIDAS models. *Energy Economics*, 78, 192-201.
- Zhang, Z., He, M., Zhang, Y., & Wang, Y. (2022). Geopolitical risk trends and crude oil price predictability. *Energy*, 258, 124824.